Mitosis notes

Mitosis is the name for the cell division that is responsible for growth, repair and replacement within living organisms, as well as during asexual reproduction. All the **somatic** cells in an organism are produced by mitosis. Only the sex cells or **gametes** are produced by the process of meiosis.

When a cell undergoes mitosis it splits in to two **daughter cells** which are identical to the original cell and to each other.

Mitosis occurs in a series of phases or stages:

Interphase

This is sometimes described as the resting phase, but it is also the stage when the chromosomes are copied. At the start the chromosomes appear as threadlike coils (**chromatin**) but gradually take on a distinct rod like structure. By the end of interphase the cell has two copies of each chromosome and each chromosome and its copy pair up and form **sister chromatids**.

Prophase

Cell division starts; centrioles or poles appear and begin to move to opposite ends of the cell. Spindle fibres form between the poles.

Metaphase

The chromatids (pairs of chromosomes) line up in the centre of the cell and attach to the spindle fibres.

Anaphase

The spindle fibres contract and pull the chromatids apart, so one copy is at each end of the cell.

Telophase

Two new nuclei form (as two nuclear membranes are formed). The chromosomes appear threadlike again (chromatin).

Cytokinesis

The cell membrane moves inwards and two new daughter cells are formed, each with its own nucleus and identical chromosomes.

Questions

1. Suggest one advantage and one genetic disadvantage of asexual reproduction.

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2. What is meant by the term somatic cells?

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3. Explain why mitosis is not used to produce gametes for sexual reproduction.

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4. What is meant by the term daughter cells?

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5. Describe the possible effect on the daughter cells if anaphase did not take place.

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